Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A charge of 1 coulomb is placed at one end of a non-conducting rod of length $0.6\ \mathrm{m}$. The rod is rotated in a vertical plane about a horizontal axis passing through the other end of the rod with angular frequency $10^4\pi$ radian/sec. Find the magnetic field at a point on the axis of rotation at a distance of $0.8\ \mathrm{m}$ from the centre of the path.
$\left(\mu_0 = 4\pi \times 10^{-7}\ \mathrm{V-s/A-m}\right)$.

(A) 1.13×10^{-3} tesla

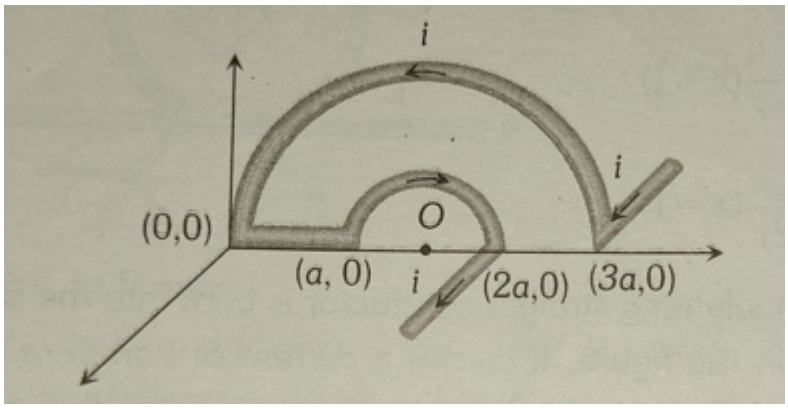
(B) 3.26×10^{-3} tesla.

(C) 1.26×10^{-3} tesla.

(D) 4.26×10^{-3} tesla.

Q.2

In the given figure net magnetic field at O will be



(A) $\frac{2\mu_0 i}{3\pi a}\sqrt{4-\pi^2}$

(B) $\frac{\mu_0 i}{3\pi a}\sqrt{4+\pi^2}$

(C) $\frac{2\mu_0 i}{3\pi a^2}\sqrt{4+\pi^2}$

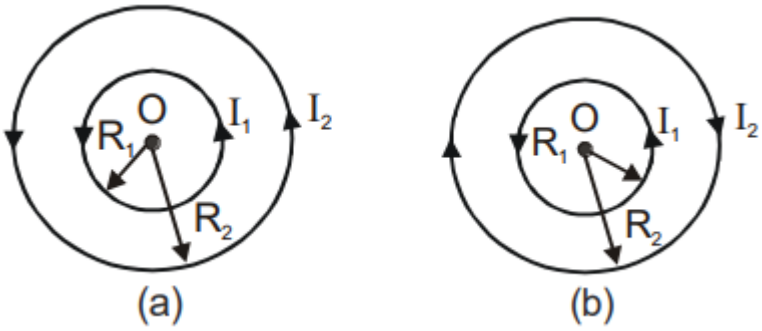
(D) $\frac{\mu_0 i}{3\pi a}\sqrt{4-\pi^2}$

Q.3

Consider two concentric coils, radii R_1 and R_2 number of turns N_1, N_2 carrying currents I_1 and I_2 as shown in fig. (a) and (b). The magnetic field at the centre when $\mathrm{I_1 = I_2 = I}$ and $\mathrm{N_1 = N_2 = N}$ in these figures will be

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respectively-



(A) $\frac{\mu_0 NI(R_1+R_2)}{2R_1R_2}, \frac{\mu_0 NI(R_2-R_1)}{2R_1R_2}$

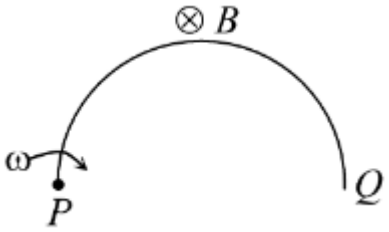
(B) $0, \frac{\mu_0 NI(R_2-R_1)}{2R_1R_2}$

(C) $\frac{\mu_0 NI(R_2-R_1)}{2R_1R_2}, 0$

(D) 0,0

Q.4

A wire is bent to form a semi-circle of radius a. The wire rotates about its one end with angular velocity ω. Axis of rotation being perpendicular to plane of the semicircle. In the space, a uniform magnetic field of induction B exists along the axis of rotation as shown. The correct statement is



(A) potential difference between P and Q is equal to $2B\omega a^2$

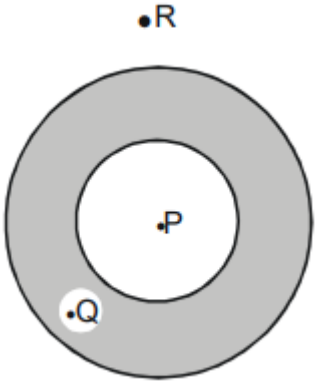
(B) potential difference between P and Q is equal to $2\pi^2 B\omega a^2$

(C) P is at higher potential than Q

(D) potential difference between P and Q is zero.

Q.5

Current is flowing through a conducting hollow pipe whose area of cross-section is shown as. The value of magnetic induction will be zero at-



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(A) Point P, Q and R

(B) Point R but not at P and Q

(C) Q but not at P and R

(D) P but not at Q and R

Q.6

A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction, extending from $x = a$ to $x = b$. The minimum value of v required so that the particle can just enter the region $x > b$ is

(A) qbB/m

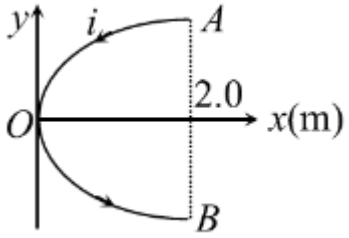
(B) $q(b-a)B/m$

(C) qaB/m

(D) $q(b+a)B/2m$

Q.7

A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current $i = 2\ \mathrm{A}$ as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ tesla. The magnetic force on the wire is



(A) $-16\hat{i}$

(B) $32\hat{i}$

(C) $-32\hat{i}$

(D) $16\hat{i}$

Q.8

A solenoid of length $0.4\ \mathrm{m}$ and having 500 turns of wire carries a current of 3 amp. a thin coil having 10 turns of wire and of radius $0.01\ \mathrm{m}$ carries a current of 0.4 amp. The torque required to hold the coil in the middle of the solenoid with its axis perpendicular to the axis of the solenoid. $\left(\mu_0 = 4\pi \times 10^{-7}\ \mathrm{V-s/A-m}\right)$ will be (in Nm)$-$

(A) 59.2×10^{-6}

(B) 5.92×10^{-6}

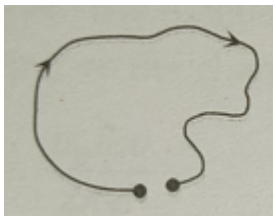
(C) 0.592×-10^{-6}

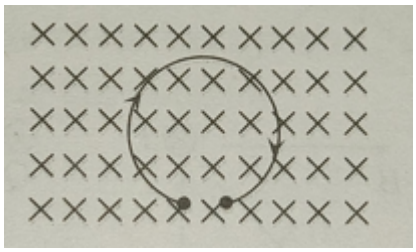
(D) 0.592×10^{-4}

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Q.9

A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in the figure. When system is put in a uniform magnetic field of strength B going into the plane of paper, the wire takes the shape of a circle. The tension in the wire is





(A) IBL

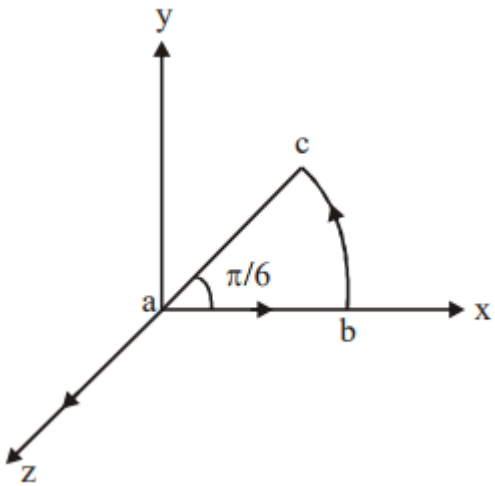
(B) $\frac{IBL}{\pi}$

(C) $\frac{IBL}{2\pi}$

(D) $\frac{IBL}{4\pi}$

Q.10

The pie-shaped current loop shown in figure subtends an angle of $\pi/6\mathrm{rad}$ and lies in the xy-plane. The radius $\mathrm{R} = 40.0\ \mathrm{cm}$. and the current $\mathrm{I} = 6.00\ \mathrm{A}$. The uniform magnetic field is parallel to the positive z-axis and has a magnitude of $0.750\ \mathrm{T}$.



(A) The magnetic force (magnitude and direction) on the segment ab is $1.8(-\hat{\mathrm{j}})$

(B) The magnetic force on the segment bc is $1.8\left(1-\frac{\sqrt{3}}{2}\right)\hat{\mathrm{i}}+0.9\hat{\mathrm{j}}\mathrm{N}$

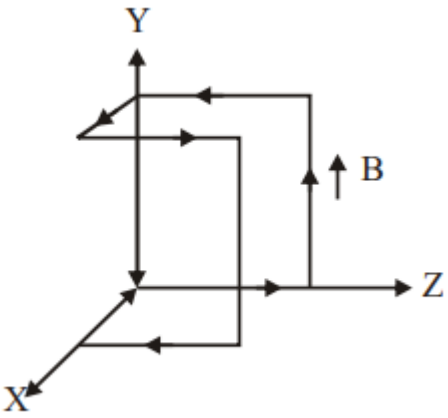
(C) The magnetic force on the segment ca is zero.

(D) The magnetic force on the segment ca is $-0.9\hat{\mathrm{i}}+0.9\sqrt{3}\hat{\mathrm{j}}$

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Q.11

Figure shows a bent coil with all edges of length $1\ \mathrm{m}$ and carrying a current of $1\ \mathrm{A}$.



(A) The magnetic moment of the loop is $2\hat{\mathrm{i}}\mathrm{Am}^2$

(B) The torque acting on the loop is $\mathrm{B}\hat{\mathrm{j}}\ \mathrm{Nm}$

(C) The magnetic moment of the loop is $\hat{i}\mathrm{Am}^2$

(D) The torque acting on the loop is $\mathrm{B}\hat{\mathrm{k}}\ \mathrm{Nm}$

Q.12

Passage

Statement:

Magnetic force on a charged particle is given by $\overrightarrow{\mathrm{F}}_{\mathrm{m}} = \mathrm{q}(\overrightarrow{\mathrm{v}} \times \overrightarrow{\mathrm{B}})$ and electrostatic force $\overrightarrow{\mathrm{F}}_{\mathrm{e}} = \mathrm{q}\overrightarrow{\mathrm{E}}$.

Question:

A particle having charge $\mathrm{q} = 1\mathrm{C}$ and mass $\mathrm{m} = 1\ \mathrm{kg}$ is released from rest at origin. There are electric and magnetic fields given by : $\overrightarrow{\mathrm{E}} = (10\hat{\mathrm{i}})\mathrm{N/C}$ for $\mathrm{x} < 1.8\ \mathrm{m}$ and $\overrightarrow{\mathrm{B}} = (-5\hat{\mathrm{k}})\mathrm{T}$ for $1.8\ \mathrm{m} \leq \mathrm{x} \leq 2.4\ \mathrm{m}$

Question

The speed with which the particle will collide the screen (in $\mathrm{m/s}$) if a screen is placed parallel to $y - z$ plane at $x = 3$ m. Neglect gravity

(A) 3

(B) 6

(C) 9

(D) 12

Q.13

Passage

Statement:

Magnetic force on a charged particle is given by $\overrightarrow{\mathrm{F}}_{\mathrm{m}} = \mathrm{q}(\overrightarrow{\mathrm{v}} \times \overrightarrow{\mathrm{B}})$ and electrostatic force $\overrightarrow{\mathrm{F}}_{\mathrm{e}} = \mathrm{q}\overrightarrow{\mathrm{E}}$.

Question:

A particle having charge $\mathrm{q} = 1\mathrm{C}$ and mass $\mathrm{m} = 1\ \mathrm{kg}$ is released from rest at origin. There are electric and

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magnetic fields given by : $\overrightarrow{E} = (10\hat{i})N/C$ for $x < 1.8$ m and $\overrightarrow{B} = (-5\hat{k})T$ for $1.8 \text{ m} \leq x \leq 2.4$ m

Question

y-coordinate of particle where it collides with the screen is ..... m. A screen is placed parallel to y-z plane at x=3 m. Neglect gravity

(A) $\frac{0.6(\sqrt{3}-1)}{\sqrt{3}}$

(B) $\frac{0.6(\sqrt{3}+1)}{\sqrt{3}}$

(C) $\frac{1.2(2-\sqrt{3})}{2}$

(D) $\frac{1.2(\sqrt{3}-1)}{\sqrt{3}}$

Q.14

Passage

Statement:

Magnetic force on a charged particle is given by $\overrightarrow{F}_m = q(\overrightarrow{v} \times \overrightarrow{B})$ and electrostatic force $\overrightarrow{F}_e = q\overrightarrow{E}$.

Question:

A particle having charge $q = 1C$ and mass $m = 1$ kg is released from rest at origin. There are electric and magnetic fields given by : $\overrightarrow{E} = (10\hat{i})N/C$ for $x < 1.8$ m and $\overrightarrow{B} = (-5\hat{k})T$ for $1.8 \text{ m} \leq x \leq 2.4$ m

Question

Time after which the particle will collide the screen is ..... second. A screen is placed parallel to $y - z$ plane at $x = 3$ m. Neglect gravity

(A) $\frac{1}{5}\left(3 + \frac{\pi}{6}\right)$

(B) $\frac{1}{5}\left(6 + \frac{\pi}{3}\right)$

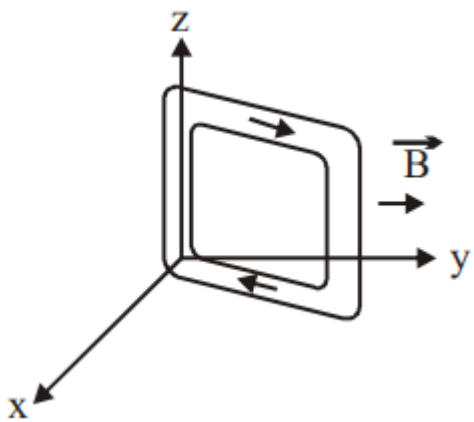
(C) $\frac{1}{3}\left(5 + \frac{\pi}{6}\right)$

(D) $\frac{1}{3}\left(6 + \frac{\pi}{18}\right)$

Q.15

Passage

The square loop in figure has sides of length 20 cm. It has 5 turns and carries a current of 2 A. The normal to the loop is at 37° to a uniform field $\overrightarrow{B} = 0.5\hat{j}T$.



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Question

Find the magnetic moment of the loop.

(A) $-0.24\hat{\mathrm{i}}+0.32\hat{\mathrm{j}}\mathrm{Am}^2$

(B) $0.24\hat{\mathrm{i}}+0.32\hat{\mathrm{j}}\mathrm{Am}^2$

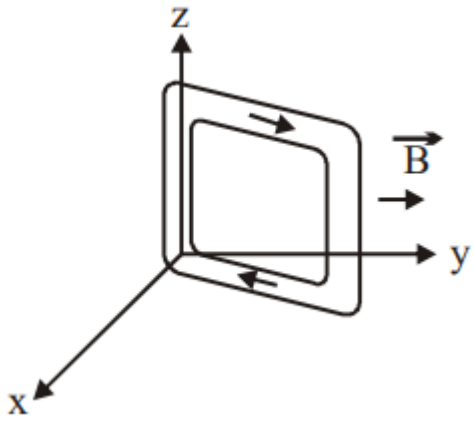
(C) $0.24\hat{\mathrm{i}}-0.24\hat{\mathrm{j}}\mathrm{Am}^2$

(D) None of these

Q.16

Passage

The square loop in figure has sides of length $20\ \mathrm{cm}$. It has 5 turns and carries a current of $2\ \mathrm{A}$. The normal to the loop is at 37° to a uniform field $\overrightarrow{\mathrm{B}}=0.5\hat{\mathrm{j}}\mathrm{T}$.



Question

Find the torque on the loop.

(A) $-0.12\hat{k}\mathrm{Nm}$

(B) $-0.12\hat{j}\mathrm{Nm}$

(C) $-0.12\hat{i}\mathrm{Nm}$

(D) $-0.06\hat{k}\mathrm{Nm}$

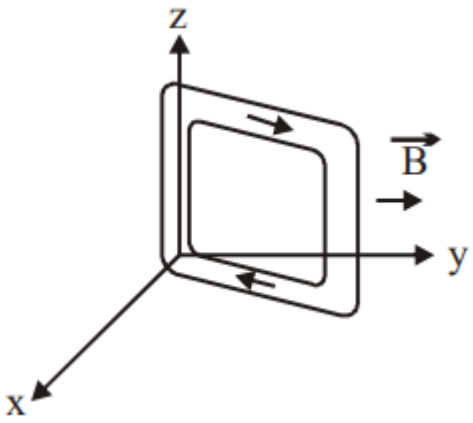
Q.17

Passage

The square loop in figure has sides of length $20\ \mathrm{cm}$. It has 5 turns and carries a current of $2\ \mathrm{A}$. The normal to the

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loop is at 37° to a uniform field $\vec{B} = 0.5\hat{j}T$.



Question

Find the work needed to rotate the loop from its position of minimum energy to the given orientation.

(A) 0.01 J

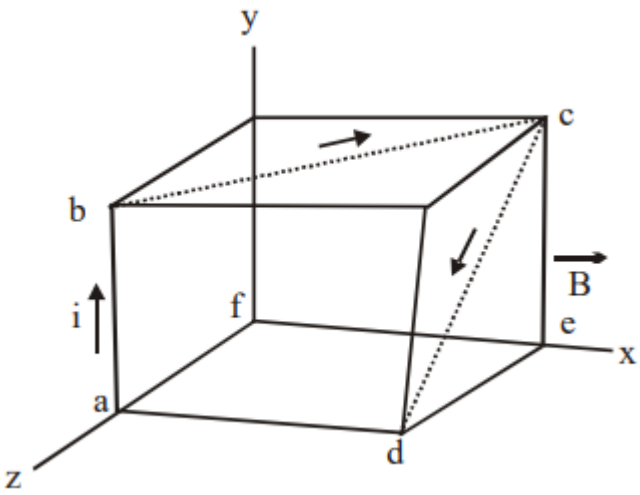
(B) 0.02 J

(C) 0.04 J

(D) 0.07 J

Q.18

The cube as shown in figure, 75.0 cm. on a side, is in a uniform magnetic field of 0.860 T parallel to the x-axis. The wire abcdef carries a current of 6.58 A in the direction indicated.



Column I	Column II
(A) The force acting on the segment ab is	(p) $(-4.24\text{N})\,\hat{k}$
(B) The force acting on the segment bc is	(q) $(-4.24\text{ N})\,\hat{j}$
(C) The force acting on the segment cd is	(r) $(4.24\text{N})\,[\hat{j}+\hat{k}]$
(D) The force acting on the segment de is	(s) zero
	(t) $(-4.24\text{N})\,\hat{i}$

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(A) $(\mathrm{A}) - \mathrm{p}, (\mathrm{B}) - \mathrm{q}, (\mathrm{C}) - \mathrm{r}, (\mathrm{D}) - \mathrm{q}$

(B) $(\mathrm{A}) - \mathrm{s}, \mathrm{t} (\mathrm{B}) - \mathrm{p}, (\mathrm{C}) - \mathrm{s}, (\mathrm{D}) - \mathrm{q}$

(C) $(\mathrm{A}) - \mathrm{t}, (\mathrm{B}) - \mathrm{r}, (\mathrm{C}) - \mathrm{q}, (\mathrm{D}) - \mathrm{s}$

(D) $(\mathrm{A}) - \mathrm{s}, (\mathrm{B}) - \mathrm{q}, (\mathrm{C}) - \mathrm{r}, (\mathrm{D}) - \mathrm{p}$

Q.19

A system consists of two parallel planes carrying currents producing a uniform magnetic field of induction B between the planes. Outside this space there is no magnetic field. The magnetic force acting per unit area of each plane is found to be $B^2/N\mu_0$. Find N

Q.20

A solenoid $60\ \mathrm{cm}$ long and of radius $4.0\ \mathrm{cm}$ has 3 layers of windings of 300 turns each. A $2.0\ \mathrm{cm}$ long wire of mass $2.5\ \mathrm{g}$ lies inside the solenoid near its centre normal to its axis; both the wire and the axis of the solenoid are in the horizontal plane. The wire is connected through two leads parallel to the axis of the solenoid to an external battery which supplies a current of $6.0\ \mathrm{A}$ in the wire. What value of current (with appropriate sense of circulation) in the windings of the solenoid can support the weight of the wire $\left(g = 9.8\ \mathrm{m/s^2}\right)$

Answers & Solutions

Q.1 Answer:

—

Solution:

The rotating rod is a current-loop whose radius a $= 0.6$ m. The magnetic field due to this currentloop at a point on its axis at a distance x from its centre is given by

$$B = \frac{\mu_0 i a^2}{2\left(a^2 + x^2\right)^{3/2}}$$

Let T be the period of rotation of the rod. Then

$$\mathrm{i} = \frac{\mathrm{q}}{\mathrm{T}} = \frac{\mathrm{q}\omega}{2\pi} = \frac{1\ \text{colulomb}\ \times 10^4 \pi/\mathrm{sec}}{2\pi} = 5 \times 10^3 \mathrm{amp}$$

Now, $\mathrm{a} = 0.6\ \mathrm{m}, \mathrm{x} = 0.8\ \mathrm{m}$ **and** $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{V - s/A - m}$

Substituting these values in eq. (i) we get

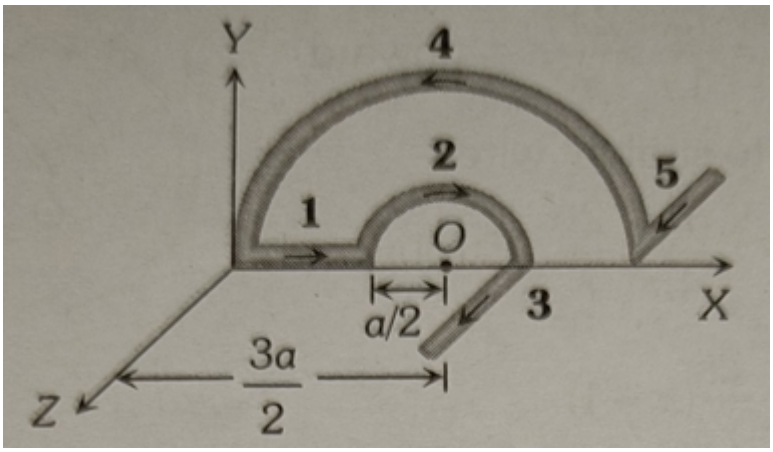
$$\mathrm{B} = \frac{\left(4\pi \times 10^{-7}\ \mathrm{V - s/A - m}\right)\left(5 \times 10^3\ \mathrm{A}\right)(0.6\ \mathrm{m})^2}{2(0.36 + 0.64)^{3/2}\ \mathrm{m}^3} = 0.36\pi \times 10^{-3} = 1.13 \times 10^{-3}\ \text{tesla}$$

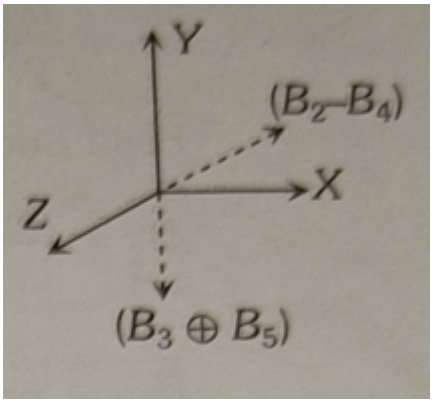
Q.2 Answer:

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$\frac{\mu_0 i}{3\pi a}\sqrt{4+\pi^2}$

Solution:





Inner loop,

$$B_1 = \frac{\mu_0}{4\pi}\frac{(\pi i)}{(a/2)} \otimes \qquad \text{(-ve z-axis)}$$

Outer loop,

$$B_2 = \frac{\mu_0}{4\pi}\left(\frac{\pi i}{3a/2}\right) \odot \qquad (+\text{ve } z\text{-axis})$$

Along z-axis,

$$B_1 - B_2 = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{a}\left[2 - \frac{2}{3}\right] \otimes$$

Wire -

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Inner,

$$B_3 = \frac{\mu_0}{4\pi} \cdot \frac{i}{(a/2)}$$

(-ve y-axis)

Outer,

$$B_4 = \frac{\mu_0}{4\pi}\left(\frac{i}{3a/2}\right) \quad \text{(-ve } y\text{-axis)}$$

$$B_3 + B_4 = \frac{\mu_0}{4\pi} \cdot \frac{I}{a}\left(2 + \frac{2}{3}\right) \quad \text{(-ve } y\text{-axis)}$$

$$B_{\text{net}} = \sqrt{(B_1 - B_2)^2 + (B_3 + B_4)^2} = \frac{\mu_0 i}{3\pi a}\sqrt{\pi^2 + 4}.$$

Q.3 Answer:

$\frac{\mu_0 \mathrm{NI}(\mathrm{R}_1+\mathrm{R}_2)}{2\mathrm{R}_1\mathrm{R}_2}, \frac{\mu_0 \mathrm{NI}(\mathrm{R}_2-\mathrm{R}_1)}{2\mathrm{R}_1\mathrm{R}_2}$

Solution:

(a) Fields due to both coils are in the same direction $\mathrm{B} = \frac{\mu_0\, \mathrm{N}_1\mathrm{I}_1}{2\mathrm{R}_1} + \frac{\mu_0\, \mathrm{N}_2\mathrm{I}_2}{2\mathrm{R}_2}$

$$\text{If } \mathrm{I}_1 = \mathrm{I}_2 = \mathrm{I}, \mathrm{N}_1 = \mathrm{N}_2 = \mathrm{N},$$
$$\mathrm{B} = \frac{\mu_0 \mathrm{NI}(\mathrm{R}_1+\mathrm{R}_2)}{2\mathrm{R}_1\mathrm{R}_2}$$

(b) Fields due to the two coils are in opposite direction, thus, total field

$$\mathrm{B} = \frac{\mu_0\, \mathrm{N}_1\mathrm{I}_1}{2\mathrm{R}_1} - \frac{\mu_0\, \mathrm{N}_2\mathrm{I}_2}{2\mathrm{R}_2}$$
$$\text{If } \mathrm{I}_1 = \mathrm{I}_2 = \mathrm{I}, \mathrm{N}_1 = \mathrm{N}_2 = \mathrm{N}$$
$$\mathrm{B} = \frac{\mu_0\, \mathrm{N}_1\mathrm{I}_1(\mathrm{R}_2-\mathrm{R}_1)}{2\mathrm{R}_1\mathrm{R}_2}$$

Q.4 Answer:

potential difference between P and Q is equal to $2B\omega a^2$

Solution:

$$E = \frac{1}{2}Bl^2\omega = \frac{1}{2}B(2a)^2\omega = 2B\omega a^2$$

Q.5 Answer:

P but not at Q and R

Solution:

Applying ampere's law at P, Q and R respectively, we find that there is no current enclosed by the circle of P. So magnetic induction at P is zero while that at Q and R is non- zero.

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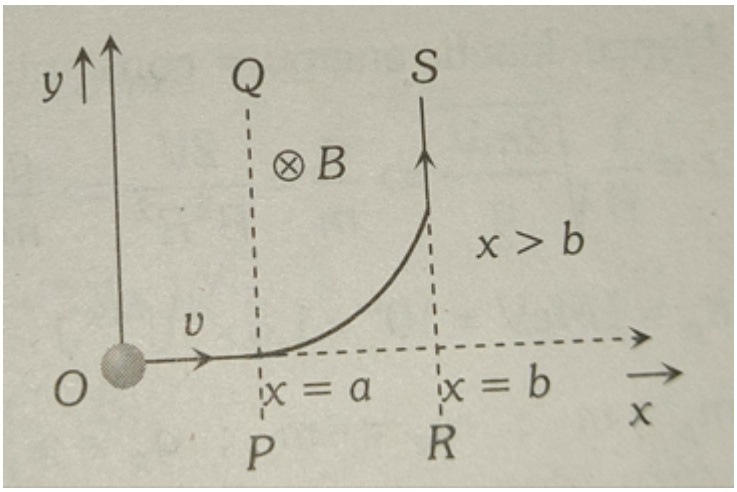
Q.6 Answer:

$q(b-a)B/m$

Solution:

Width of magnet field region $(b-a) \leq R$, where R is its radius of curvature inside magnetic field

$$R = \frac{mv}{qB} \geq (b-a) \Rightarrow v_{\min} = \frac{(b-a)qB}{m}$$



Q.7 Answer:

$32\hat{i}$

Solution:

$$F_m = 2[4(-\hat{j}) \times 4(-\hat{k})] = F_m = 32\hat{i}$$

Q.8 Answer:

5.92×10^{-6}

Solution:

The magnetic field at the centre of the solenoid is

$$\mathrm{B} = \mu_0 \mathrm{ni}\left(4\pi \times 10^{-7}\right) \times (500/0.4) \times 3 \ \mathrm{N/A.m.}$$

The torque acting on a current-carrying coil having N turns (say), placed perpendicular to the axis at the centre of the solenoid is-

$$\begin{aligned} \tau = \ \mathrm{BiNA} \ &= \left(4\pi \times 10^{-7}\right) \times (500/0.4) \times 3 \times 0.4 \times 10 \times \pi(0.01)^2 \\ &= 6\pi^2 \times 10^{-7} = 5.92 \times 10^{-6} \ \mathrm{N.m.} \end{aligned}$$

Q.9 Answer:

$\frac{IBL}{2\pi}$

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Solution:

Let an elemental length d subtending an angle $d\theta$ at the centre of the circle

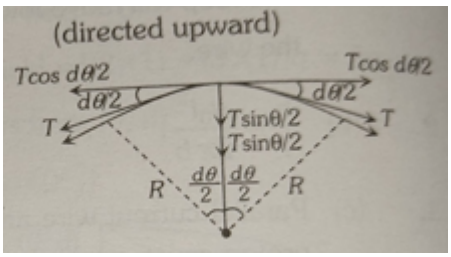
Force $F_B = BI(dl)$

$$F_B = BI(Rd\theta)$$
$$= BI\left(\frac{L}{2\pi}\right)d\theta$$

Now, $2T\sin\frac{d\theta}{2}$ add up to balance F_B

At equilibrium -

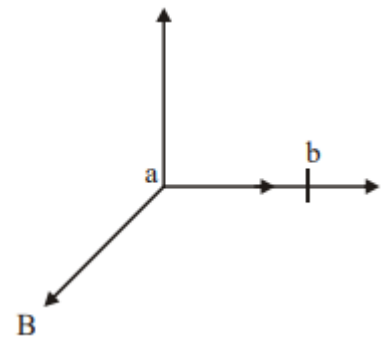
$$2T\sin\frac{d\theta}{2} = BI\frac{L}{2\pi}d\theta$$
$$\Rightarrow 2T\frac{d\theta}{2} = BI\frac{L}{2\pi}d\theta \quad \left[\text{For small angle } \sin\frac{d\theta}{2} \approx \frac{d\theta}{2}\right.$$
$$\Rightarrow T = \frac{BIL}{2\pi}.$$

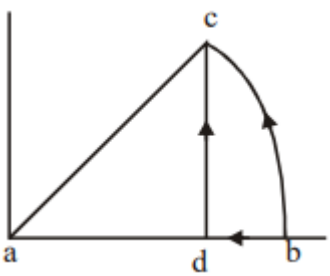


Q.10 Answer:

The magnetic force (magnitude and direction) on the segment ab is $1.8(-\hat{j})$

Solution:





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(A) $\overrightarrow{B} = 0.75(\hat{k})$

$F_{ab} = B\ell I = 0.75 \times 6 \times 40 \times 10^{-2}$

$F_{ab} = 1.8\ N \Rightarrow \overrightarrow{F}_{ab} = 1.8(-\hat{j})$

$\overrightarrow{F}_{bc} = \overrightarrow{F}_{bd} + \overrightarrow{F}_{dc} = BI(bd)\hat{j} + BI(dc)\hat{i}$

$= 0.75 \times 6\left[(R - R\cos 53^\circ)\,\hat{i} + R\sin 30^\circ \hat{j}\right] = 1.8\left(1 - \frac{\sqrt{3}}{2}\right)\hat{i} + 0.9\hat{j}N$

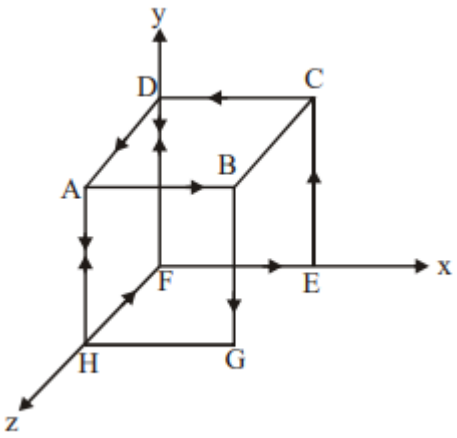
(C) $\overrightarrow{F}_{ca} = \overrightarrow{F}_{cd} + \overrightarrow{F}_{da} = -0.9\hat{i} + 0.9\sqrt{3}\hat{j}$

We can see that net force is zero on the loop.

Q.11 Answer:

The magnetic moment of the loop is $\hat{i}Am^2$

Solution:



Join additional wires at AH and DF carrying same current in opposite directions.

It will not affect the overall loop. Loop ABGH and DCEF will cancel as their magnetic moment is in opposite directions.

(A) Net magnetic moment of the loop = Magnetic moment of $ADFH = -i\ell^2\hat{i} = \hat{i}Am^2$

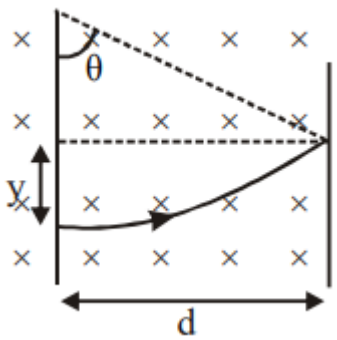
(B) $\overrightarrow{B} = B\hat{j}$

$\vec{\tau} = \overrightarrow{m} \times \overrightarrow{B} = -i\ell^2\hat{i} \times \hat{B} = -i\ell^2\, B\hat{k} = B\hat{k}Nm$

Q.12 Answer:

6

Solution:



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In magnetic field speed remain same

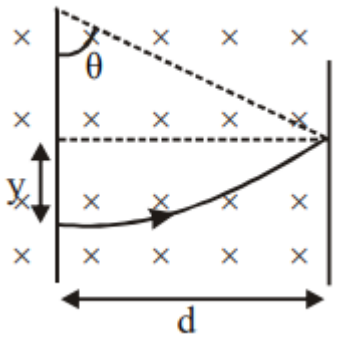
In $EF, v^2 = 0 + 2ax$

$= 0 + 2 \times \frac{10 \times 1}{1} \times 1.8 \Rightarrow \mathrm{v} = 6\ \mathrm{m/s}$

Q.13 Answer:

$\frac{1.2(2-\sqrt{3})}{2}$

Solution:



In magnetic field speed remain same

In $EF, v^2 = 0 + 2ax$

$0 + 2 \times \frac{10 \times 1}{1} \times 1.8 \Rightarrow \mathrm{v} = 6\ \mathrm{m/s}$

In MF, charge particle will move on circular path of radius

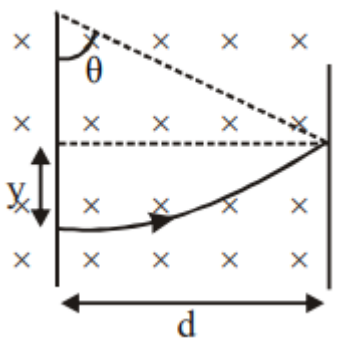
$$\mathrm{r} = \frac{\mathrm{mv}}{\mathrm{qB}} = \frac{1 \times 6}{1 \times 4}\ \mathrm{m}$$
$$\sin\theta = \frac{\mathrm{d}}{\mathrm{r}} = \frac{0.6 \times 5}{6} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$
$$\tan\theta = \frac{\mathrm{d}}{\mathrm{r}-\mathrm{y}}$$
$$\frac{1}{\sqrt{3}} = \frac{0.6}{(6/5)-\mathrm{y}}; \mathrm{y} = 1.2\left(1 - \frac{\sqrt{3}}{2}\right)$$
$$\mathrm{T} = \mathrm{t}_1 + \mathrm{t}_2$$
$$1.8 = \frac{1}{2}\frac{\mathrm{qE}}{\mathrm{m}}\mathrm{t}_1^2 \Rightarrow \mathrm{t}_1 = \frac{3}{5}$$

t_2 **time in magnetic field** $= \frac{\mathrm{T}}{2\pi}\frac{\pi}{6} = \frac{\mathrm{T}}{12} = \frac{\pi}{30} \Rightarrow \mathrm{T} = \frac{3}{5} + \frac{\pi}{30}$

Q.14 Answer:

$\frac{1}{5}\left(3 + \frac{\pi}{6}\right)$

Solution:



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In magnetic field speed remain same

In $EF, v^2 = 0 + 2ax$

$0 + 2 \times \frac{10\times1}{1} \times 1.8 \Rightarrow \mathrm{v} = 6\ \mathrm{m/s}$

In MF, charge particle will move on circular path of radius

$$\mathrm{r} = \frac{\mathrm{mv}}{\mathrm{qB}} = \frac{1\times6}{1\times4}\ \mathrm{m}$$
$$\sin\theta = \frac{\mathrm{d}}{\mathrm{r}} = \frac{0.6\times5}{6} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$
$$\tan\theta = \frac{\mathrm{d}}{\mathrm{r}-\mathrm{y}}$$
$$\frac{1}{\sqrt{3}} = \frac{0.6}{(6/5)-\mathrm{y}}; \mathrm{y} = 1.2\left(1 - \frac{\sqrt{3}}{2}\right)$$
$$\mathrm{T} = \mathrm{t}_1 + \mathrm{t}_2$$
$$1.8 = \frac{1}{2}\frac{\mathrm{qE}}{\mathrm{m}}\mathrm{t}_1^2 \Rightarrow \mathrm{t}_1 = \frac{3}{5}$$

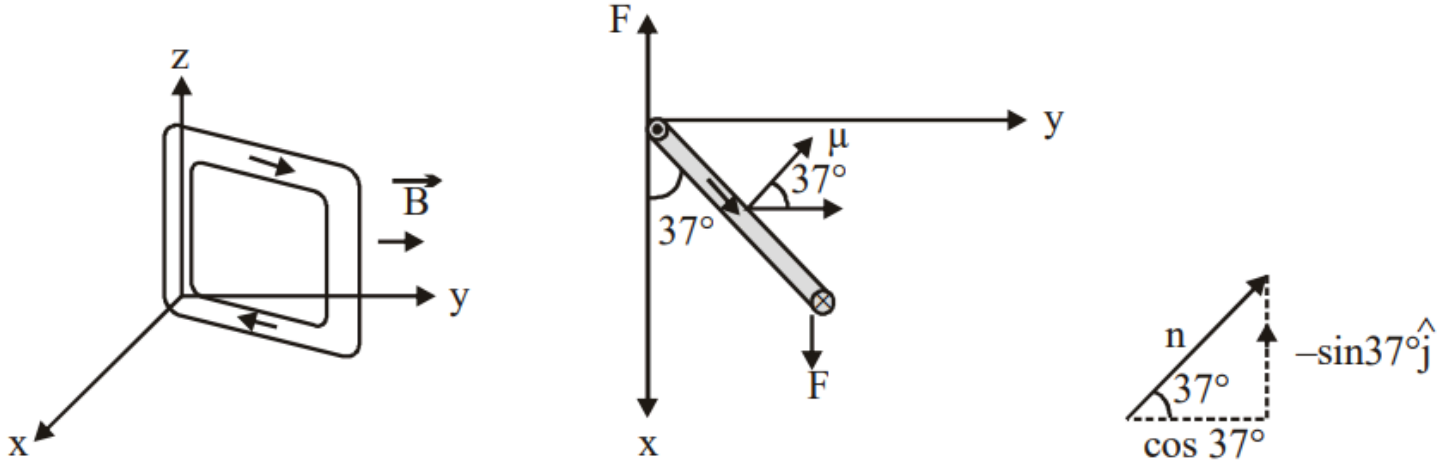
t_2 **time in magnetic field** $= \frac{\mathrm{T}}{2\pi}\frac{\pi}{6} = \frac{\mathrm{T}}{12} = \frac{\pi}{30} \Rightarrow \mathrm{T} = \frac{3}{5} + \frac{\pi}{30}$

Q.15 Answer:

$-0.24\hat{\mathrm{i}} + 0.32\hat{\mathrm{j}}\mathrm{Am}^2$

Solution:

(i) From figure we see that the unit vector normal to loop $\hat{\mathrm{n}} = -\sin 37^\circ\hat{\mathrm{i}} + \cos 37^\circ\hat{\mathrm{j}} = -0.6\hat{\mathrm{i}} + 0.8\hat{\mathrm{j}}$



The magnetic moment is $\mu = \mathrm{NIA}\,\hat{\mathrm{n}} = (5)(2)(0.2)^2(-0.6\hat{\mathrm{i}} + 0.8\hat{\mathrm{j}}) = -0.24\hat{\mathrm{i}} + 0.32\hat{\mathrm{j}}\mathrm{Am}^2$

(ii) The torque, $\tau = \mu \times \overrightarrow{\mathrm{B}} = (-0.24\hat{\mathrm{i}} + 0.32\hat{\mathrm{j}}) \times (0.5\hat{\mathrm{j}}) = -0.12\hat{\mathrm{k}}\mathrm{Nm}$

(iii) The potential energy of the loop is $\mathrm{U} = -\mathrm{mB}\cos\theta$ **where** $\mu = \mathrm{NIA} = 0.4\mathrm{Am}^2$ **and the position of minimum energy is** $\theta = 0$

Thus, the external work, $\mathrm{W}_{\mathrm{ext}} = +\Delta\mathrm{U}$**, needed to rotate it to the given orientation, is given by** $\mathrm{U}_\mathrm{f} - \mathrm{U}_\mathrm{i} = (-\mu\mathrm{B}\cos 37^\circ) - (\mu\mathrm{B}\cos 0^\circ) = (0.4)(0.5)(1 - 0.8) = 0.04\ \mathrm{J}$

The external work is positive since the dipole moment is rotated away from alignment with the field.

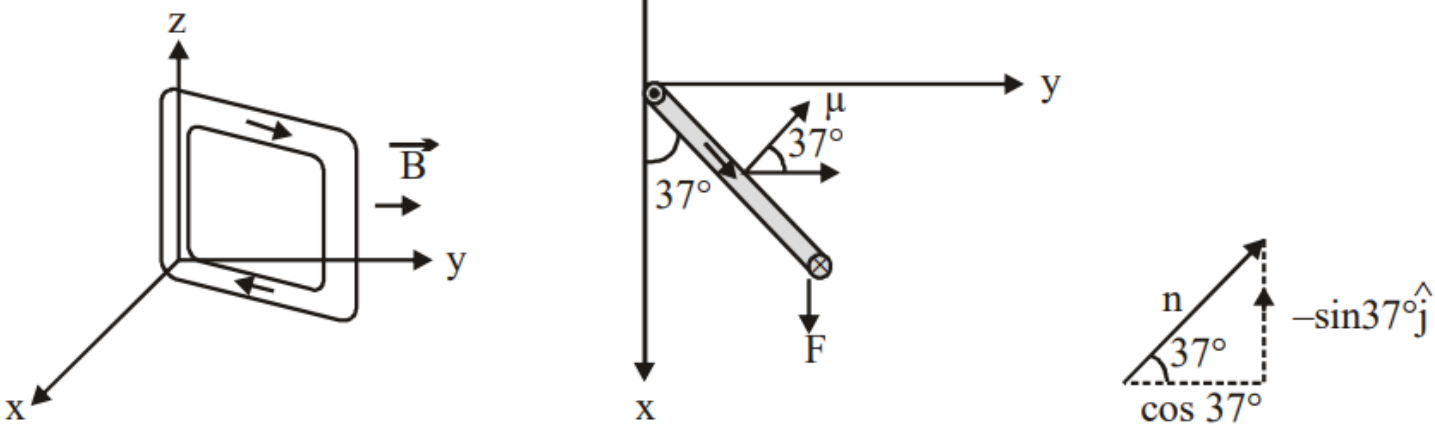
Q.16 Answer:

$-0.12\hat{k}\mathrm{Nm}$

Solution:

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(i) From figure we see that the unit vector normal to loop $\hat{n} = -\sin 37^\circ \hat{i} + \cos 37^\circ \hat{j} = -0.6\hat{i} + 0.8\hat{j}$



The magnetic moment is $\mu = \text{NIA}\,\hat{n} = (5)(2)(0.2)^2(-0.6\hat{i} + 0.8\hat{j}) = -0.24\hat{i} + 0.32\hat{j}\,\text{Am}^2$

(ii) The torque, $\tau = \mu \times \overrightarrow{B} = (-0.24\hat{i} + 0.32\hat{j}) \times (0.5\hat{j}) = -0.12\hat{k}\,\text{Nm}$

(iii) The potential energy of the loop is $U = -mB\cos\theta$ **where** $\mu = \text{NIA} = 0.4\,\text{Am}^2$ **and the position of minimum energy is** $\theta = 0$

Thus, the external work, $W_{ext} = +\Delta U$**, needed to rotate it to the given orientation, is given by** $U_f - U_i = (-\mu B\cos 37^\circ) - (\mu B\cos 0^\circ) = (0.4)(0.5)(1 - 0.8) = 0.04\ \text{J}$

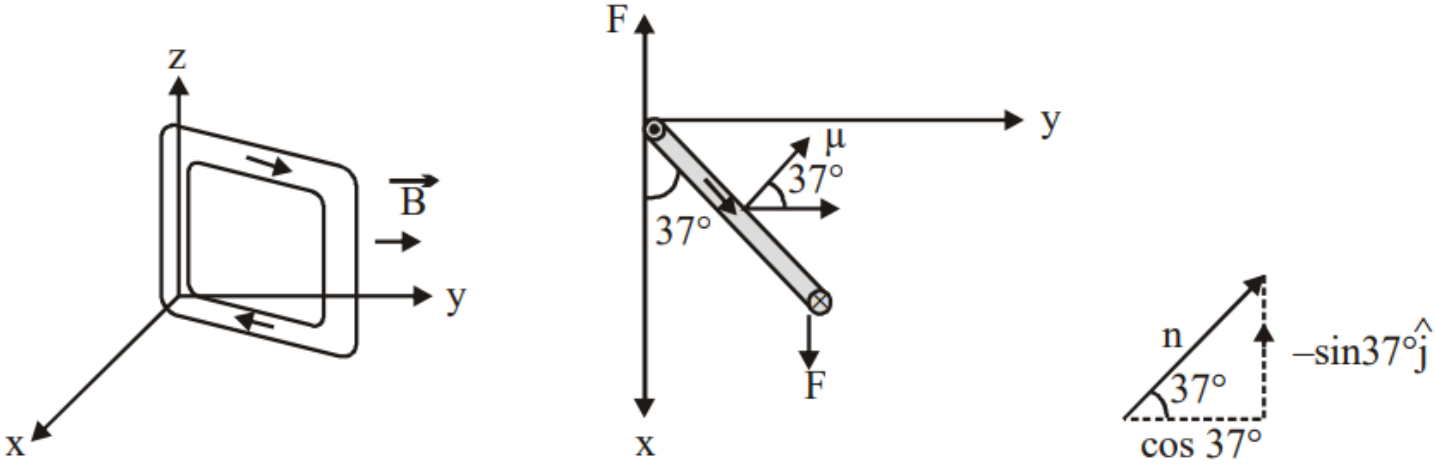
The external work is positive since the dipole moment is rotated away from alignment with the field.

Q.17 Answer:

$0.04\ \text{J}$

Solution:

(i) From figure we see that the unit vector normal to loop $\hat{n} = -\sin 37^\circ \hat{i} + \cos 37^\circ \hat{j} = -0.6\hat{i} + 0.8\hat{j}$



The magnetic moment is $\mu = \text{NIA}\,\hat{n} = (5)(2)(0.2)^2(-0.6\hat{i} + 0.8\hat{j}) = -0.24\hat{i} + 0.32\hat{j}\,\text{Am}^2$

(ii) The torque, $\tau = \mu \times \overrightarrow{B} = (-0.24\hat{i} + 0.32\hat{j}) \times (0.5\hat{j}) = -0.12\hat{k}\,\text{Nm}$

(iii) The potential energy of the loop is $U = -mB\cos\theta$ **where** $\mu = \text{NIA} = 0.4\,\text{Am}^2$ **and the position of minimum energy is** $\theta = 0$

Thus, the external work, $W_{ext} = +\Delta U$**, needed to rotate it to the given orientation, is given by** $U_f -$

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$U_i = (-\mu B \cos 37^\circ) - (\mu B \cos 0^\circ) = (0.4)(0.5)(1 - 0.8) = 0.04\ J$

The external work is positive since the dipole moment is rotated away from alignment with the field.

Q.18 Answer:

$(A) - p, (B) - q, (C) - r, (D) - q$

Solution:

$(A) - p, (B) - q, (C) - r, (D) - q$

(A) $\overrightarrow{F} = \overrightarrow{I}_{ab} \times \overrightarrow{B} = I(\ell_{ab} B)\hat{j} \times \hat{i} = -(6.58\ A)(0.750\ m)(0.860\ T)\hat{k} = (-4.24\ N)\hat{k}$

(B) $\overrightarrow{F} = I\vec{\ell}_{bc} \times \overrightarrow{B} = I(\ell_{bc} B)\left[\frac{(\hat{i}-\hat{k})}{\sqrt{2}} \times \hat{i}\right] = -(6.58\ A)(0.750\ m)(0.860\ T)\hat{j} = (-4.24\ N)\hat{j}$

(C) $\overrightarrow{F} = I\vec{\ell}_{cd} \times \overrightarrow{B} = I(\ell_{cd} B)\left[\frac{(\hat{k}-\hat{j})}{\sqrt{2}} \times \hat{i}\right] = -(6.58\ A)(0.750\ m)(0.860\ T)[\hat{j} + \hat{k}]$

$\Rightarrow \quad \overrightarrow{F} = (4.24\ N)[\hat{j} + \hat{k}]$

(D) $\overrightarrow{F} = I\vec{\ell}_{de} \times \overrightarrow{B} = I\ell_{de} B[-\hat{k} \times \hat{i}] = -(6.58\ A)(0.750\ m)(0.860\ T)\hat{j} = (-4.24\ N)\hat{j}$

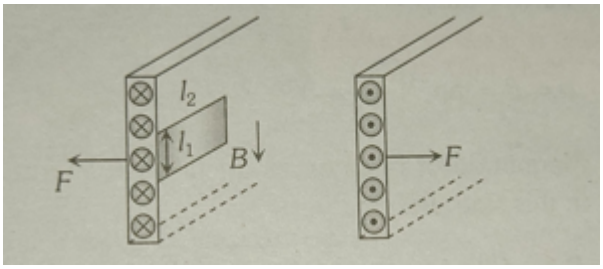
Q.19 Answer:

—

Solution:

Let J is the current per unit length.

$$B = 2\left[\frac{\mu_0 J}{2}\right] = \mu_0 J$$



Magnetic field due to one plane $= B/2$

Force on other plane $F = \frac{B}{2}(Jl_1)\, l_2$

This force is on the shaded part.

Force per unit area : $\frac{F}{l_1 I_2} = \frac{B}{2} J = \frac{B}{2}\frac{B}{\mu_0} = \frac{B^2}{2\mu_0}$

Hence $N = 2$.

Q.20 Answer:

—

Solution:

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Number of turns per unit length of the solenoid,

$n = \frac{\text{Total number of turns}}{\text{Length of the solenoid}} = \frac{3 \times 300}{0.60 \text{ m}} = 1500 \text{ m}^{-1}$

If I be the current through the solenoid, magnetic field inside the solenoid, $B = \mu_0 n I$

B acts along the axis of the solenoid.

As the wire carrying current I' lies inside the solenoid near its centre and normal to its axis, magnetic force acting on the wire, i.e.,

$F = I' l B = I' l \ \mu_0 n I$

Since both the wire and the axis of the solenoid are in the horizontal plane, F acts in the vertical direction. Further, with appropriate sense of circulation of current, F can be made to act vertically upwards.

Thus, for the wire to be supported

$F = mg$ **(weight of the wire)**

or $I' \mid \mu_0 n I = mg$

$$\Rightarrow I = \frac{mg}{I' l \mu_0 n}$$

$$I = \frac{\left(2.5 \times 10^{-3}\right)(9.8)}{6\,(2.0 \times 10^{-2})\,(4\pi \times 10^{-7})\,1500} \text{ A}$$

$$I = 108 \text{ A}$$

(as $m = 2.5 \text{ g} = 2.5 \times 10^{-3} \text{ kg}, I' = 6 \text{ A}, l = 2.0 \times 10^{-2} \text{ m}$**).**

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